

A large, thick, black L-shaped frame composed of two perpendicular bars. One bar is vertical on the left side, and the other is horizontal at the bottom, forming an open square shape that frames the central text.

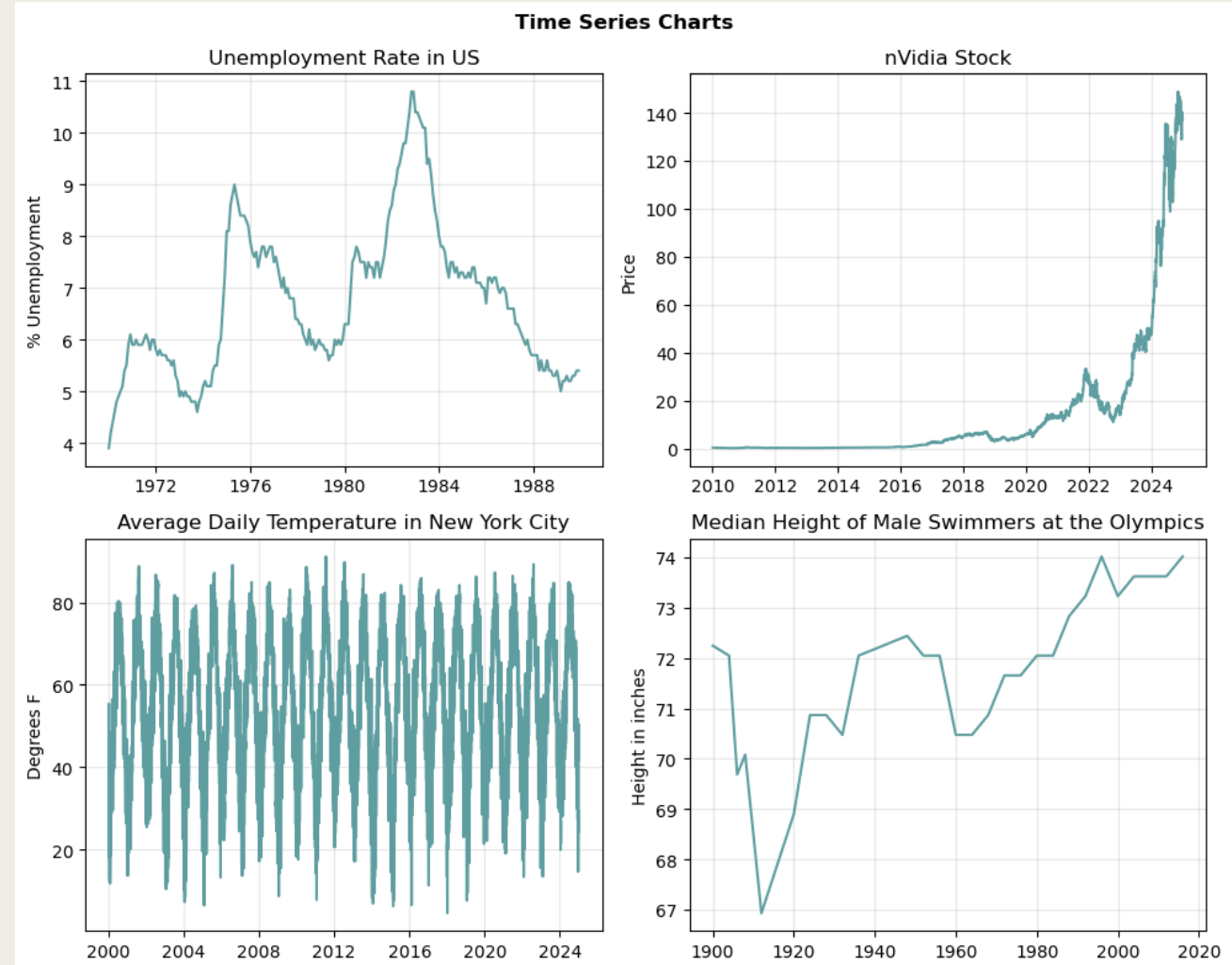
TIME SERIES

Outline

- Examine time-series data
- Explore and visualize time-series data
- Review basic forecasting methods
- Explore exponential smoothing
- Examine ARIMA models
- Construct forecasts using simple methods, exponential smoothing, and ARIMA.

Time Series

- A series of data in order of time
- A lot of data around us is measured over a span of time
 - *Stock prices*
 - *Federal funds rate*
 - *Sales*
 - *Weather*
- Time-series data is indexed on time.



Uses

- Forecasting
- Anomaly detection. E.g., outages, fraud
- Causal impact/intervention analysis: Effect of promotions, policy changes, shocks.
- Signal decomposition & features: Extract trend/seasonality for modeling

Common Applications

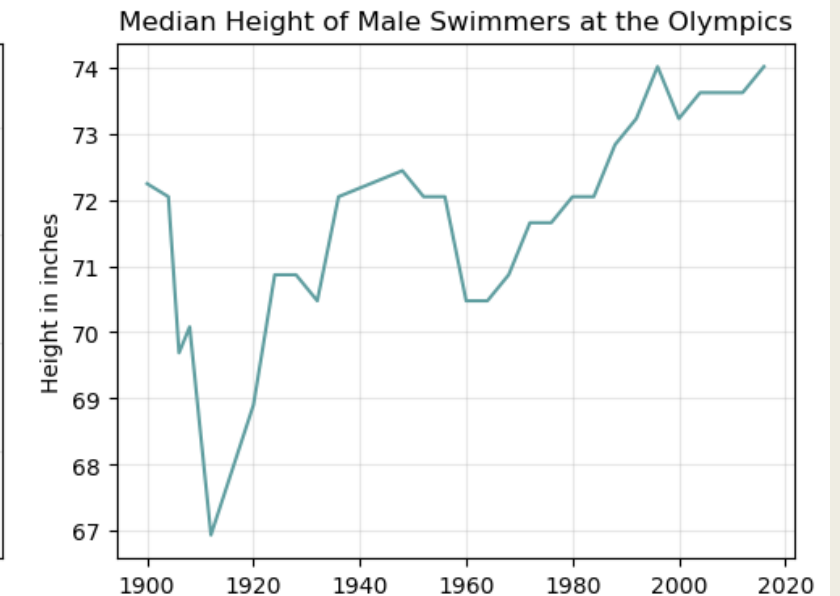
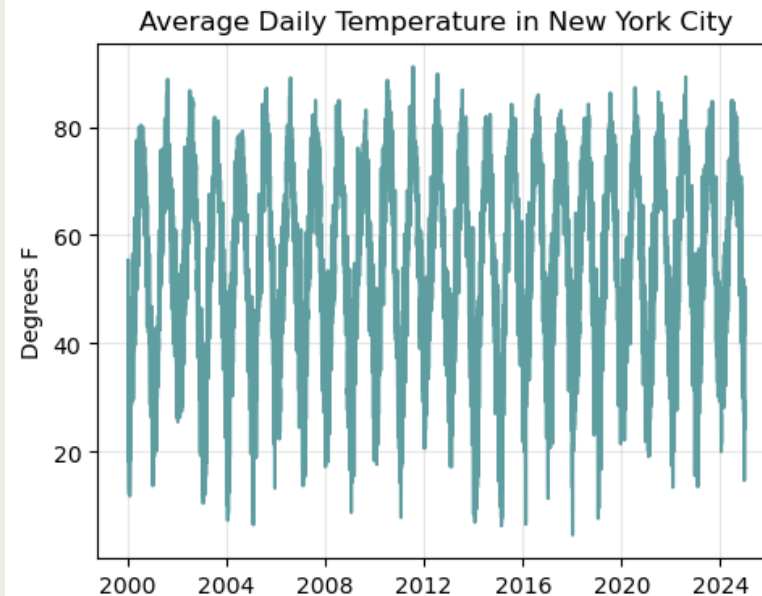
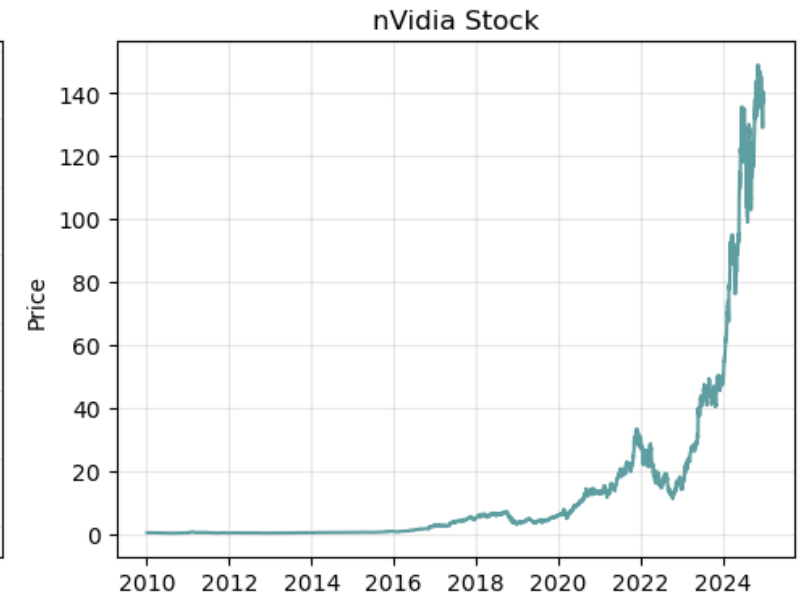
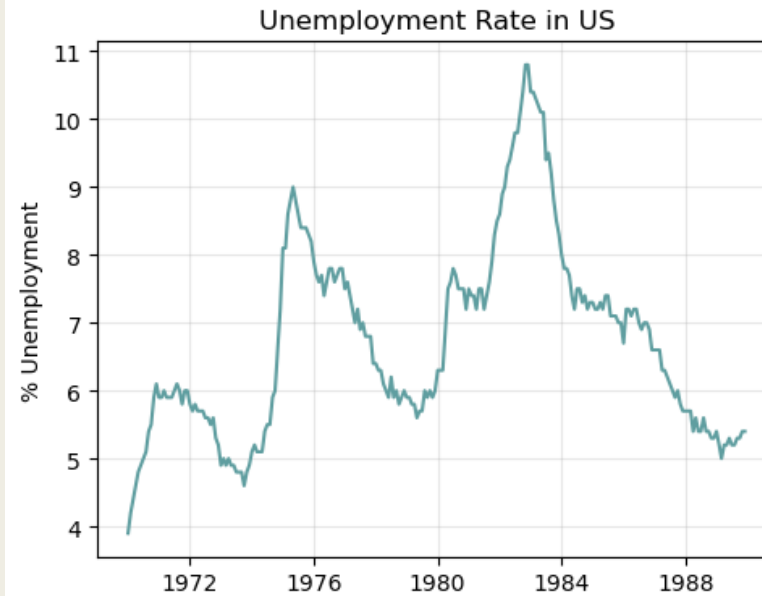
- Price forecasting: Equities, commodities, interest rates
- Demand & capacity planning: Retail SKUs, call centers
- Load & generation forecasting: Electricity demand, wind/solar output
- Traffic & travel time prediction (highways, public transit)
- Healthcare monitoring: ECG/ICU vitals, disease incidence
- Marketing analytics: Effect of promotions, churn or engagement trajectories, dynamic pricing

Simplifying Assumptions

- We make some simplifying assumptions about time-series
 - 1. Consecutive observations are equally spaced
 - 2. Apply a discrete-time index (even though this may only hold approximately)
 - *E.g., daily stock returns are only available for weekdays, not weekends and holidays.*

*Can you
identify a
pattern in
these time
series?*

Time Series Charts



Time Series Patterns

- Trend

- *A long-term increase or decrease in the data. Could be linear, non-linear, and may go from an increasing trend to a decreasing trend*

- Seasonal

- *Pattern repeats itself every season. Here season may be period of the year or day of the week. Seasonality is always of a fixed or known frequency*

- Cycle

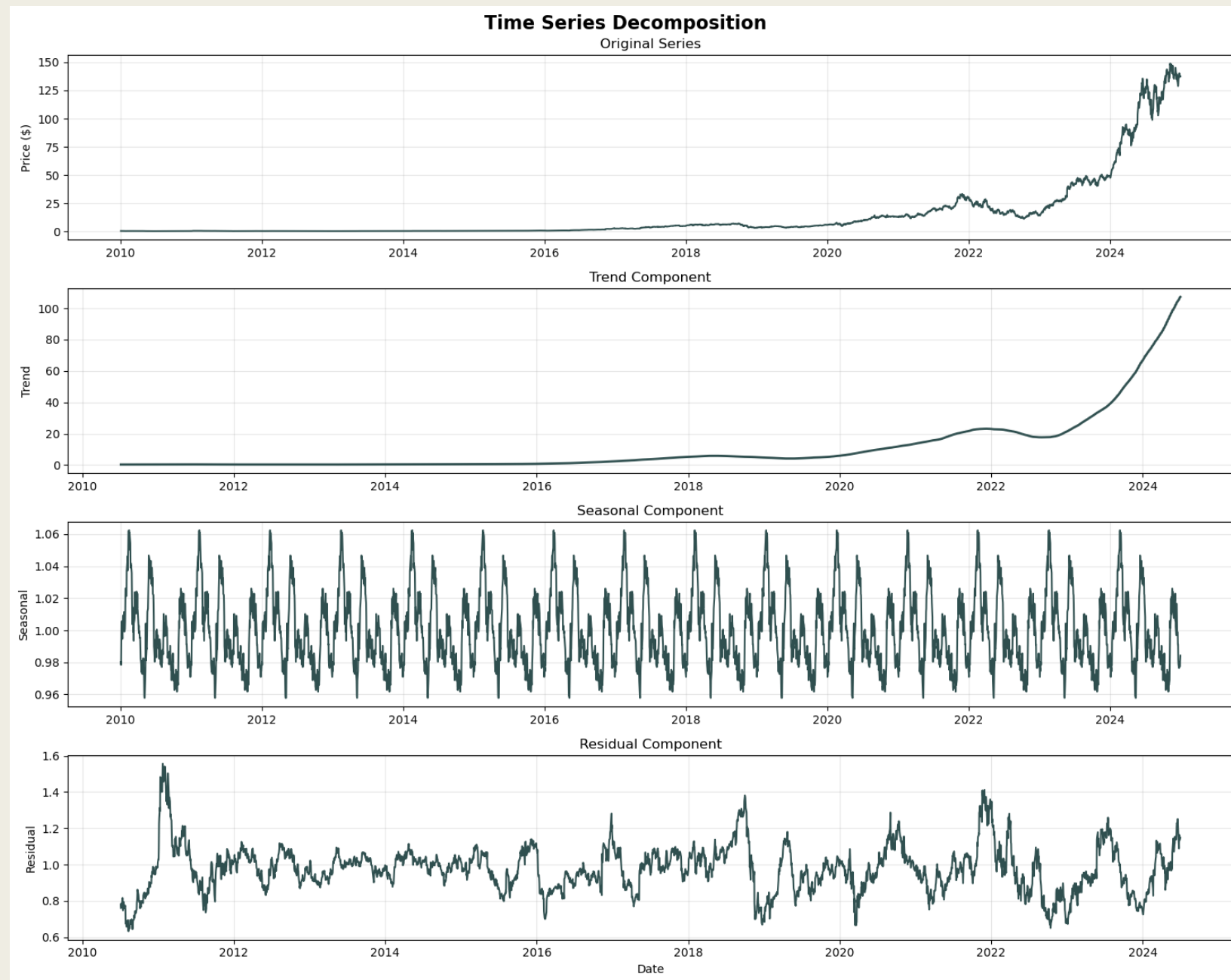
- *Data exhibit rises or falls that are not of a fixed frequency*

- On the other hand, a time-series that does not exhibit any pattern is a Stationary Process. White Noise is the simplest Stationary Process.

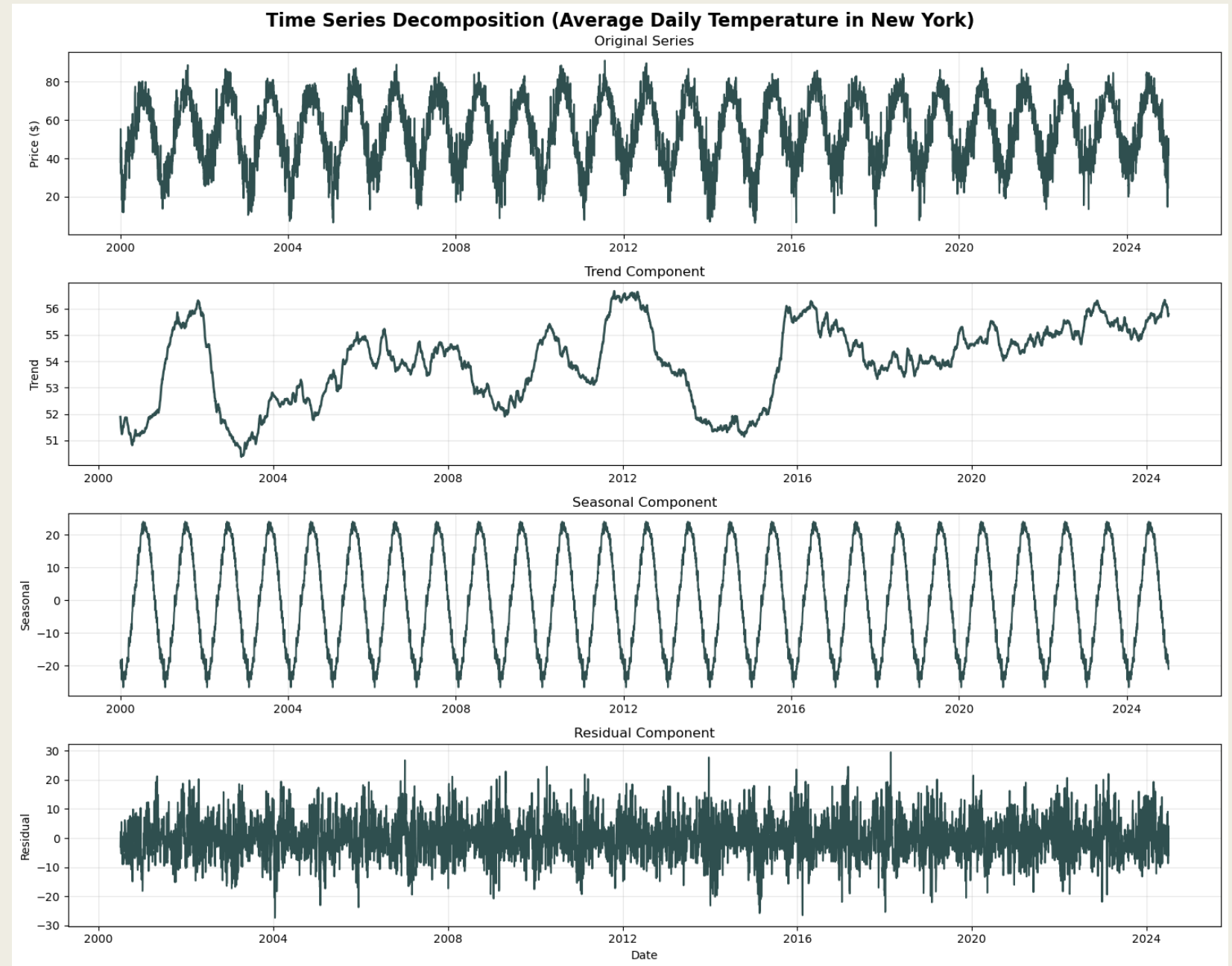
Decomposing a Time Series

- Time-series may have multiple components in varying strengths. To better understand the series it is useful to divide it into its components: :
 - *Trend components*
 - *Seasonal component*
 - *Remainder*
 - $y_t = T_t + S_t + R_t$
- Decomposition can be used to measure the strength of trend and seasonality

Decomposing time series of nVidia stock



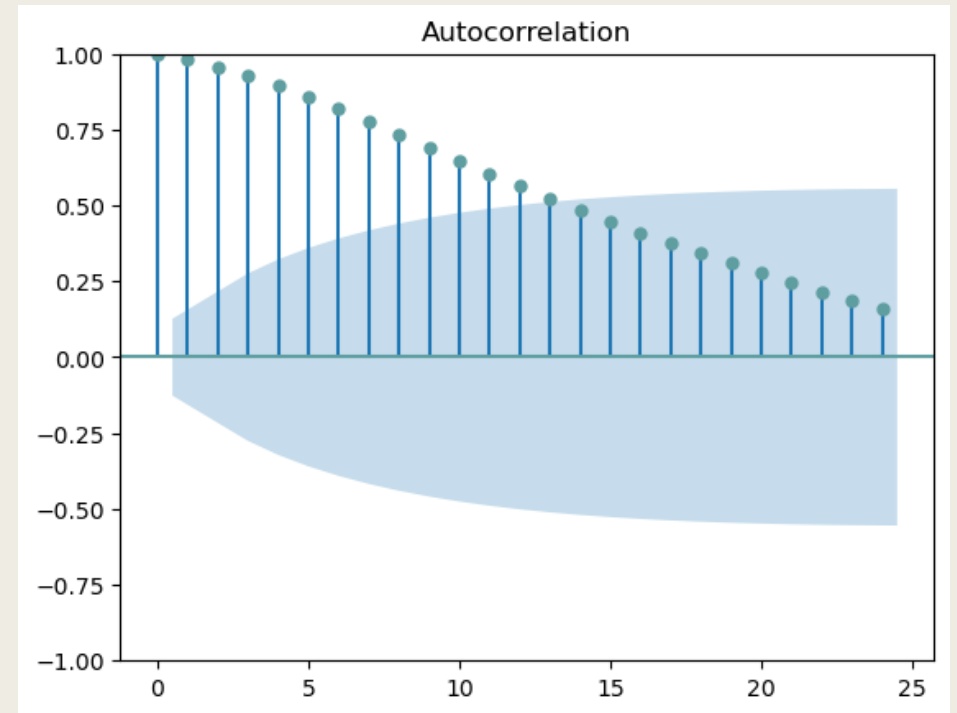
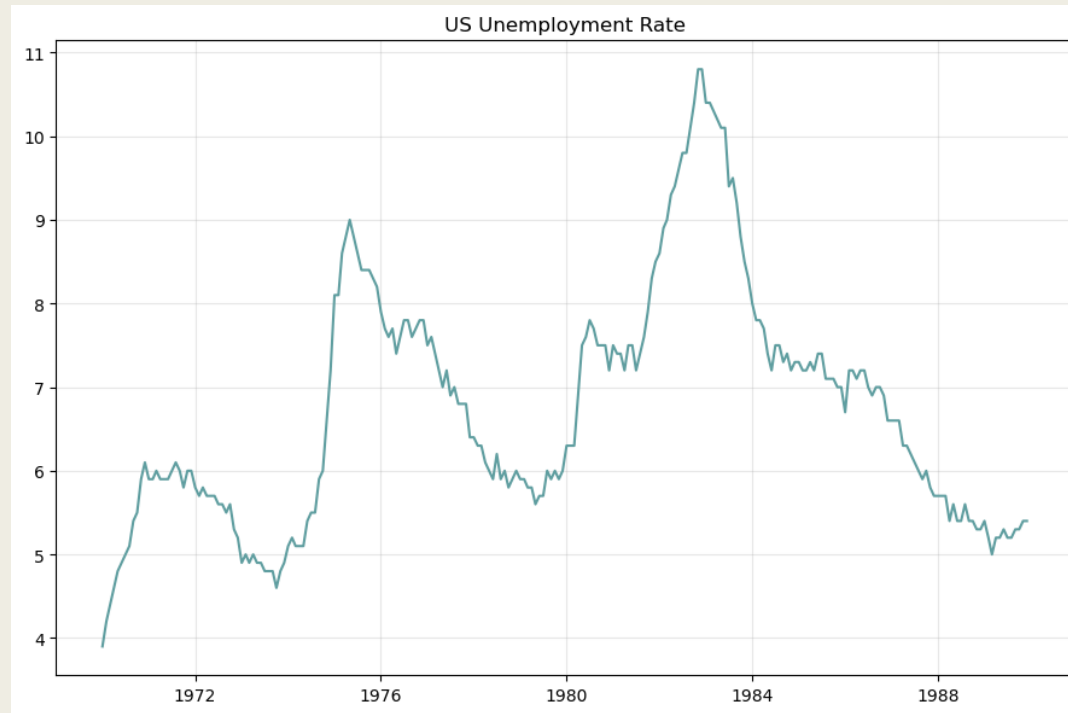
Decomposing time series of average daily temperature in New York



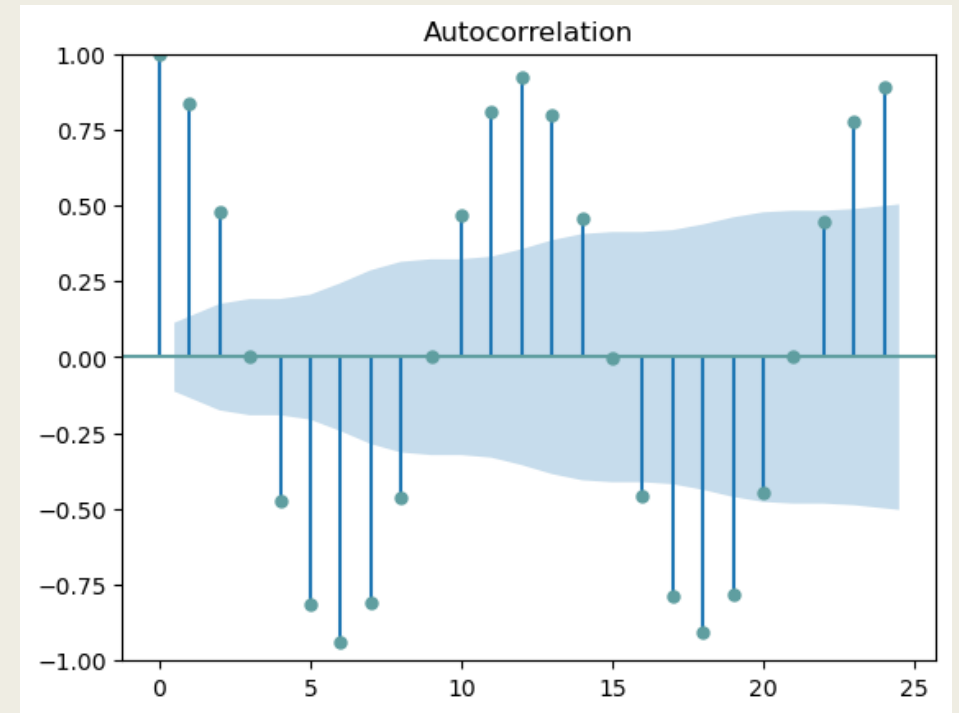
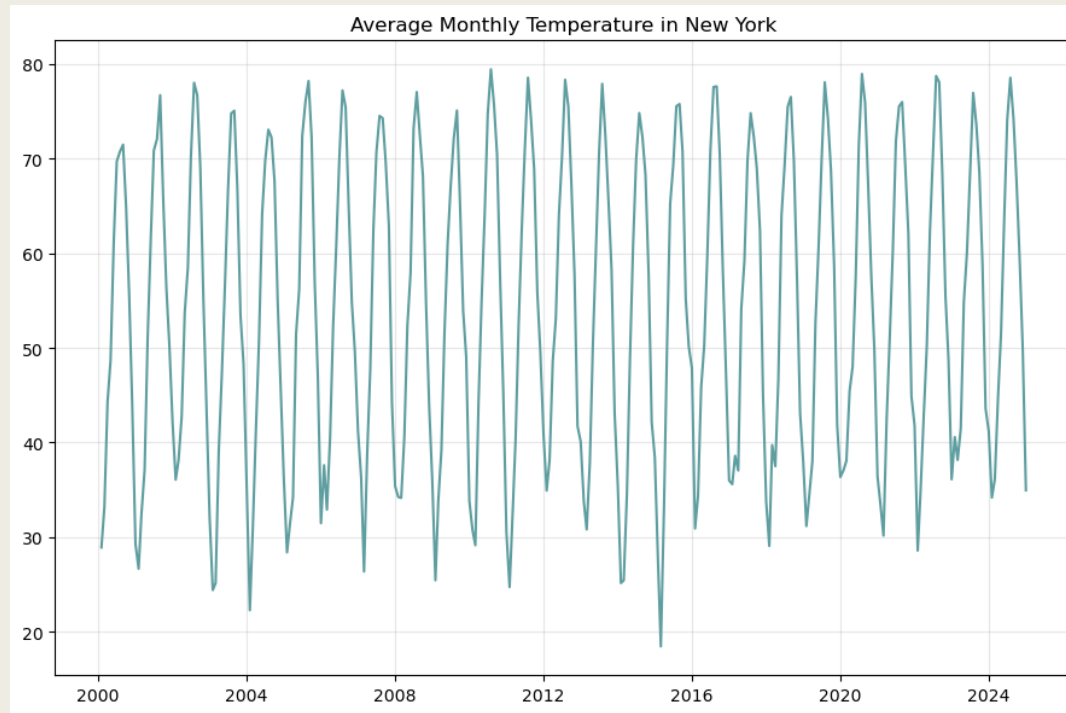
Time Series Patterns

- A common observation across time-series is that prior observations influence later observations.
- The influence may be from the previous observation (trend), an observation that is a fixed number of periods ago (seasonality), or an observation that is not a fixed number of periods ago (cycle).
- Influence of prior observations can be examined by looking at lagged correlations or autocorrelations.

Spot the Pattern: Trend, Seasonality, Cyclicality?



Spot the Pattern: Trend, Seasonality, Cyclicity?



Forecasts

- Predictions with time-series data are called forecasts
- Of course, not all forecasts are derived from time-series data. Judgmental forecasts are often used when
 - *There is no historical data*
 - *Data is incomplete or available after a delay*
- Three approaches to incorporating judgmental forecasts
 - *Rely entirely on judgment when there is no available data*
 - *When data are available, generate statistical forecasts and adjust using judgment*
 - *When data are available, generate statistical and judgmental forecasts independently and combine the two.*

Judgmental Forecasts

- Delphi method
- Forecasting by analogy
- Scenario forecasting
- New product forecasting: Sales force composite, executive opinion, customer intentions

Simple Forecasting Methods

■ Average Method

- *Uses the average of historical data as the forecast for all future points.*
- *Appropriate if the series fluctuates around a stable mean.*

■ Naïve Method

- *Projects the last observed value forward for all future periods.*
- *Useful when data shows no trend or seasonality.*
- *Since this is the best prediction for a random walk, these are also called random walk forecasts*

■ Seasonal Naïve Method

- *Forecast is equal to the last observed value from the same season*

■ Drift Method

- *Extends a linear trend based on the first and last observed values in training. Captures long-term upward or downward drift.*
- *Allow forecasts to increase or decrease over time, where the amount of change over time (called the drift) is set to be average change seen in historical data*



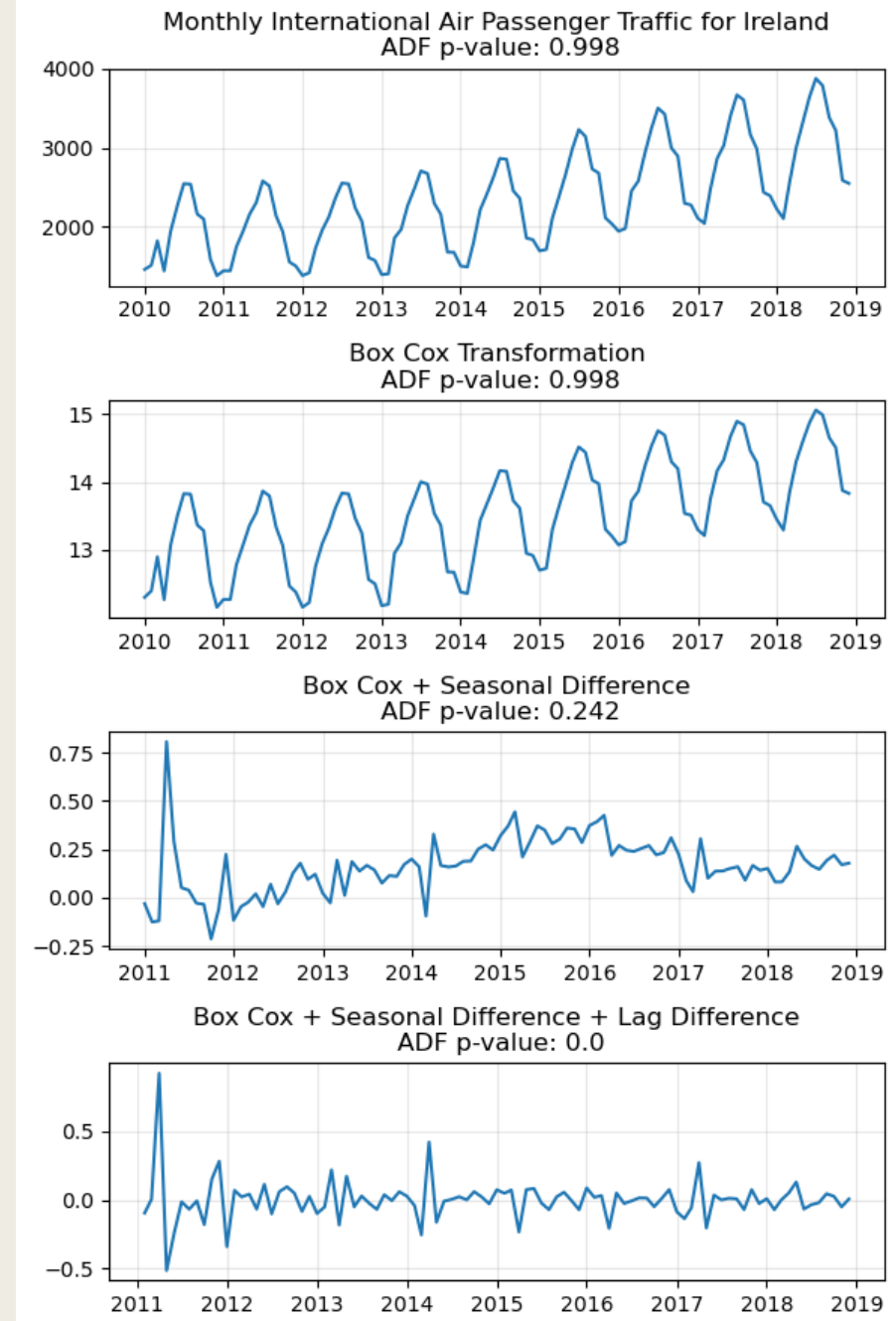
ARIMA

Stationary Process

- Autoregressive (AR) and Moving Average (MA) models assume the series is Stationary
- Stationary Process has constant mean, variance and covariance (between identically spaced data points) over time.
- However, most time-series are non-stationary
- A natural first step to working with time series is to make it stationary. This is accomplished by
 - *Stabilizing Variance*
 - *Seasonal Differencing*
 - *Lag Differencing*
- Stationarity can be evaluated by statistical tests
 - *Augmented Dickey Fuller Test*
 - *KPSS Test*

Stationary Process

- A non-stationary process can be transformed into a (weakly) stationary process through transformations that remove trend and stabilize variance.
 - *Stabilize variance: Box-Cox Transformation*
 - *Remove seasonality and trend: Differencing*

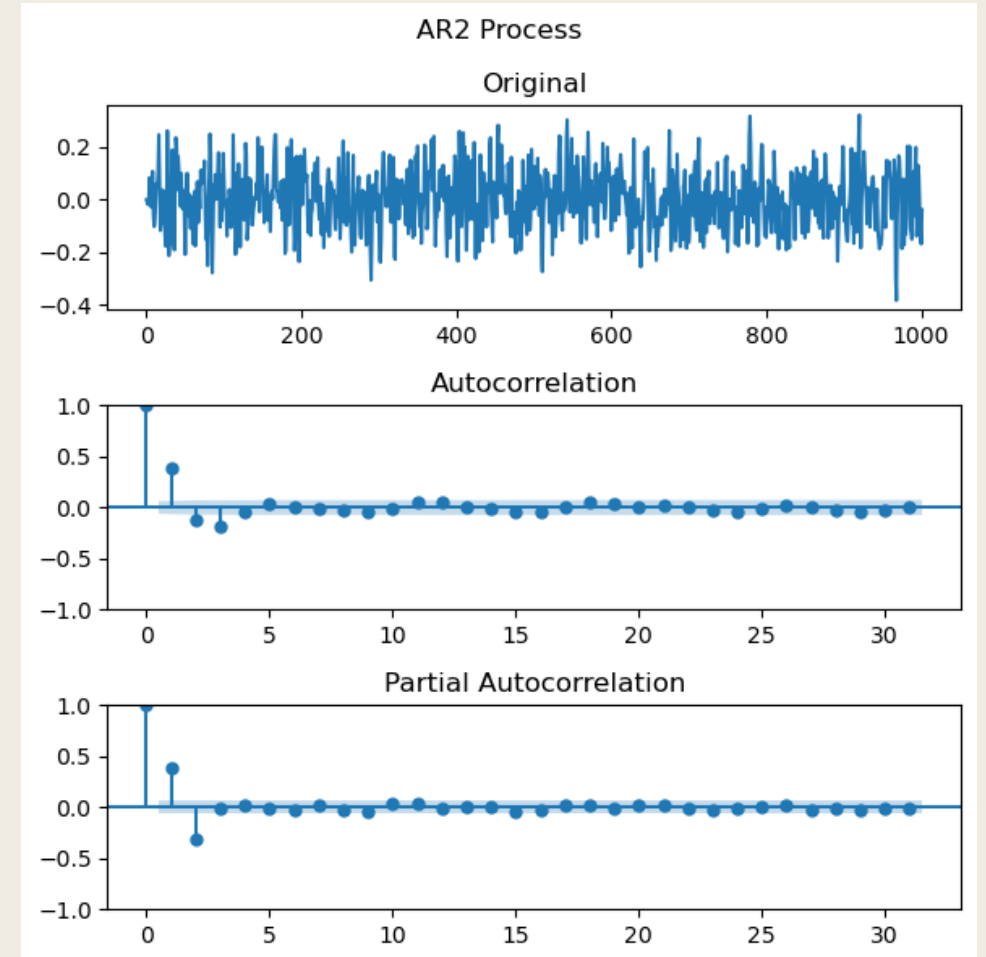


Autoregressive Models (AR)

- Forecast of a variable depends linearly on its own previous values. In other words, it is a regression of the variable against itself.
- Denoted as AR(p)
- Modelled as linear combination of constant C, present error, and past values of the series, AR order, p, determines the number of past values
- $y_t = c + \varphi_1 y_{t-1} + \varphi_2 y_{t-2} + \varphi_3 y_{t-3} + \dots + \varepsilon_t$
 - where ε_t is white noise, and c is a constant

Autoregressive Model (AR)

- In a pure $AR(p)$ process, the PACF drops sharply after lag p .
- The ACF tails off gradually.
- E.g., if PACF has significant spikes at lags 1 and 2 only it suggests an $AR(2)$ process
- In a seasonal $AR(P)$ process, look for spikes in PACF at lag s
 - where s is the seasonal period

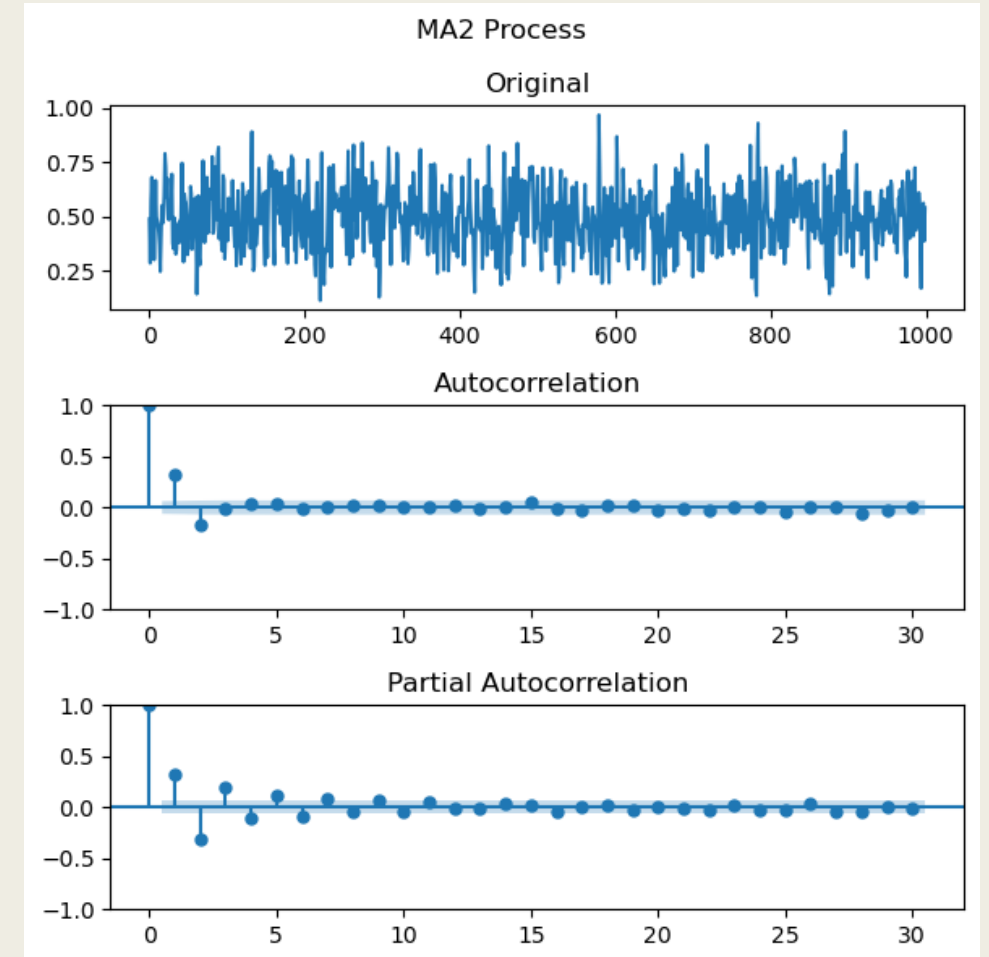


Moving Average (MA)

- Current value depends on current and past error terms
- Denoted as MA(q)
- Modelled as linear combination of series mean (μ), present error, and past errors. MA order, q , determines the number of past errors
- A moving average model uses past forecast errors in a regression-like model
- $y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q}$
 - where ε_t is white noise, and μ is the series mean

Moving Average Model (MA)

- In a pure MA(q) process, the ACF drops sharply after lag q .
- The PACF tails off gradually.
- E.g., if ACF has significant spikes at lags 1 and 2 only it suggests an MA(2) process
- In a seasonal MA(Q) process, look for spikes in ACF at lag s
 - where s is the seasonal period



ARIMA

- ARMA: Combination of AR and MA models
- ARIMA
 - $ARIMA = AR + I + MA$
 - *AutoRegressive Integrated Moving Average*
 - *ARMA after applying differencing to the series*

ARIMA

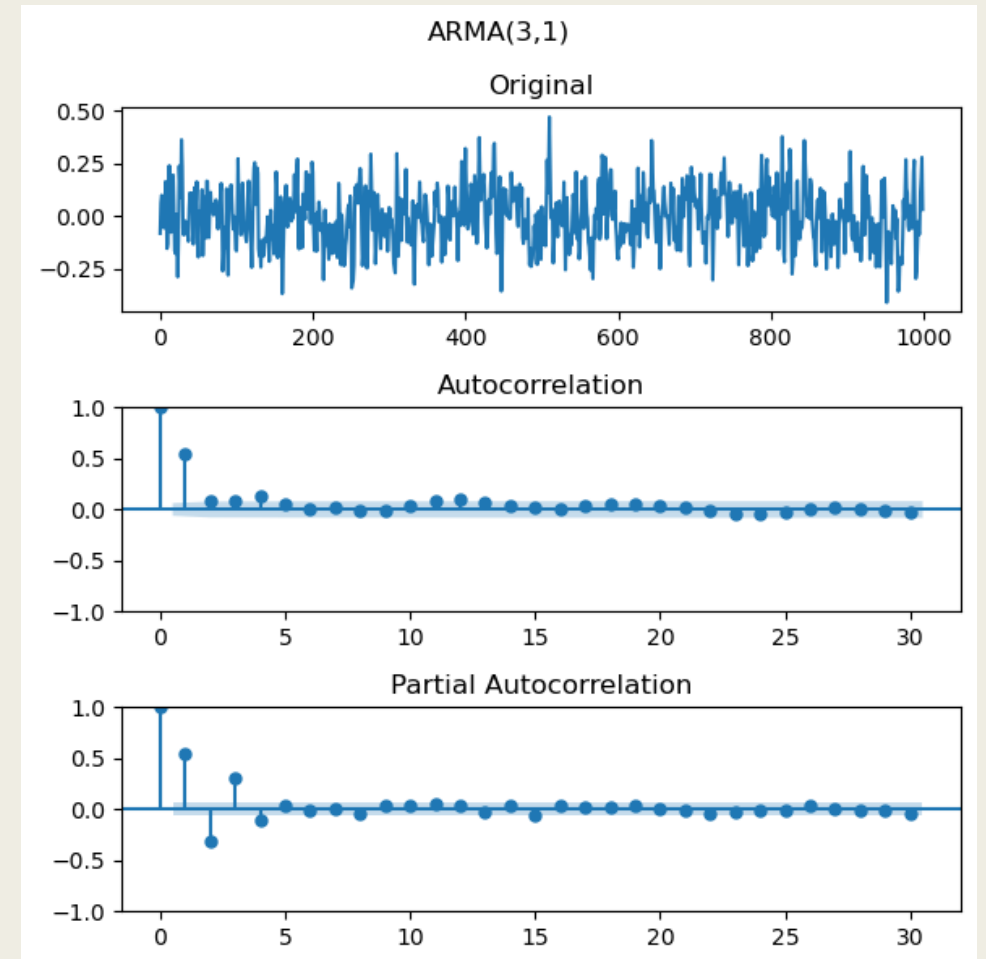
- $y'_t = c + \varphi_1 y'_{t-1} + \dots + \varphi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q} + \varepsilon_t$
 - where y'_t is the differenced series
- ARIMA(p,d,q)
 - p – order of autoregressive (AR) part
 - d – degree of first differencing
 - q – order of moving average (MA) part

SARIMA

- (Non-Seasonal) ARIMA + seasonal terms = Seasonal ARIMA
- SARIMA(p,d,q)(P,D,Q) _{m}
 - (p,d,q) is non-seasonal component
 - (P,D,Q) is seasonal component
 - m is number of periods per season
- where
 - d = Number of lag differences
 - p = Number of ordinary Autoregressive Lags
 - q = Number of ordinary Moving average lags
 - D = Number of seasonal differences
 - P = Number of seasonal Autoregressive Lags
 - Q = Number of seasonal Moving Average lags
 - m = Seasonal period or number of observations per year

SARIMA

- When both AR and MA terms are present, neither ACF nor PACF show a clear cut-off. Instead, both tail off gradually.
- In such cases, it is difficult to determine orders p and q from PACF and ACF plots. But, these plots may still offer hints or guidance on values of p and q that may be good.
- Ultimately, model parameters are determined by examining multiple combinations p , q , P and Q .
- Model selection is based on AIC, BIC and residual errors.



SARIMA Estimation

- ARIMA model is estimated using maximum likelihood estimation. Technique finds the values of the parameters which maximize the probability of obtaining the data that we have observed

Steps in Model Selection

1. Check Stationarity
2. Apply transformations, if needed
3. Examine ACF, PACF
4. Grid Search for ARIMA
5. Fit ARIMA with lowest AIC/BIC
6. Examine Residuals
7. Forecast

Model Selection Process

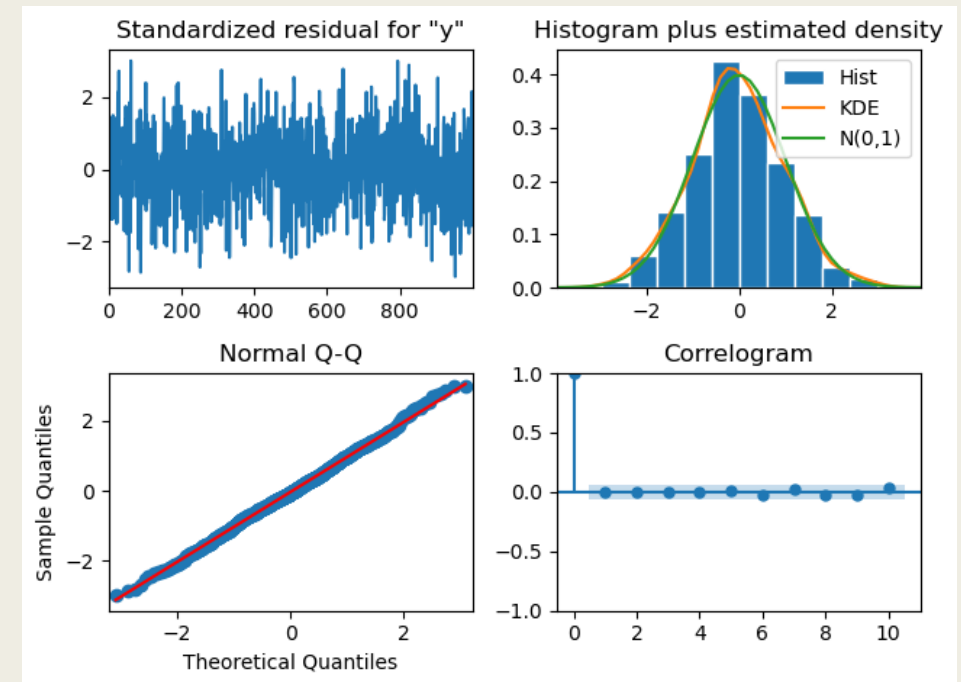
Grid Search

- Generate a list of candidate values for p, q, P, and Q.
- Create combinations of these values
- Fit SARIMA
- Collect AIC (or BIC)
 - *AIC penalizes complexity*
 - $AIC = 2k - 2\ln(\text{likelihood function})$
 - *Where k is number of parameters*
- Select model with smallest AIC (or BIC)

Model Selection Process

Fit ARIMA and Examine Residuals

- Fit ARIMA with the lowest AIC from grid search
- Now, AIC is a relative measure of model quality. Residuals are an absolute measure of model quality.
- Residual Analysis
 - *For a good model, residuals should be normally distributed and uncorrelated*
 - *QQ Plot indicates whether residuals are approximately normally distributed*
 - *Ljung Box test evaluates whether residuals are uncorrelated*



Model Selection Process

Forecast

- Forecasts for desired period or length of test
- Reverse transformations
- Evaluate forecasts
- Visualize

ADVANCED METHODS

SARIMAX: External Variables

- SARIMA relies on historical data to construct forecasts
- However, it is possible that external variables also have an impact on our time series and can therefore be good predictors of future values.

$$y_t = \text{SARIMA}(p,d,q)(P,D,Q)m + \sum_{i=1}^n \beta_i X_t^i$$

- Including external variables can be beneficial as one may find important predictors for the target. The coefficients can be used to understand sensitivity or elasticity of target to changes in exogenous variable (e.g., effect of unemployment rate on GDP).
- An important caveat is that this technique relies on availability of data on exogenous variables. This is not a problem for what-if situations (e.g., X is the price of products which the retailer can control), or if exogenous variables can be forecast.

Forecasting Multiple Time Series

VAR

- VAR is a multivariate time series model that captures the dynamic interactions between multiple variables. In VAR models, each variable is a function of
 - *Its own past values*
 - *Past values of other variables*
 - *Error terms*
- SARIMAX assumes a unidirectional relationship with the exogenous variable influencing the endogenous variable. Vector Autoregression (VAR) relaxes this constraint to capture bidirectional relationships between time series.
- For two series y_1 and y_2 , the relationship can be expressed using matrix notation as:

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} + \begin{bmatrix} \phi_{1,1} & \phi_{1,2} \\ \phi_{2,1} & \phi_{2,2} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} \epsilon_{1,t} \\ \epsilon_{2,t} \end{bmatrix}$$

Forecasting Multiple Time Series

VAR

- Granger causality tests to see if one time series does in fact influence the other(s). Despite the name of the test it only assesses predictive causality, not true causality. This is because we are only determining whether past values of a time series are statistically significant in predicting another time series. This is an important test to see if a VAR model is needed or if a SARIMA will suffice.
- VAR coefficients can shed light on the current and lagged effects of other exogenous variables.

Prophet

- Prophet is Facebook's forecasting tool designed for business time series with
 - *Trend changes and seasonality*
 - *Holiday effects*
 - *Missing data handling*
 - *Automatic parameter tuning*
- Results are easy to interpret and implementation is quick

Long Short-Term Memory (LSTM)

- LSTM networks are specialized neural networks for
 - *Sequential data processing*
 - *Long-term dependency learning*
 - *Non-linear pattern recognition*
 - *Complex temporal relationships*
- Useful when
 - *Complex non-linear patterns exist*
 - *Multiple features need to be integrated*
 - *Long-term dependencies are important*
 - *High accuracy is required*
 - *Computational resources are available*

Summary

- In this module we,
 - *examined time series data*
 - *inspected characteristics of time series*
 - *utilized basic forecasting methods*
 - *examined ARIMA family of forecasting methods*
 - *reviewed advanced forecasting methods*